

A COMPREHENSIVE APPROACH TO MRI BRAIN IMAGE SEGMENTATION AND CLASSIFICATION USING DEEP LEARNING APPROACHES

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ABSTRACT

Building an accurate and efficient model for tumor classification is challenging due to the complexities in characteristics and the variations in MRI modalities. Conventional methods not fully exploit the diverse information available from each modality. Using deep residual learning and fusion of multi modal image features would overcome limitations. The proposed approach considers the diversities in illumination across different MRI modalities, which can provide unique details about the tumor. Model architectures incorporate mechanisms for spatial and channel attention, enabling the network to concentrate on pertinent regions and features within MRI scans. Deep Learning models like EfficientNetB3 and EfficientNetV2 are utilized as base architectures for the proposed classification models and RESUNET is utilized for the proposed segmentation model. Classification models like EfficientNetB3, achieved an accuracy of nearly 99%, EfficientNetV2 achieved an accuracy of 95.2% and Vision Transformer achieved an accuracy of 95%. Segmentation models like ResUnet achieved a loss of nearly 0.2007. These impressive results demonstrate the significant advancements in image classification and segmentation tasks.

Keywords: Medical image analysis, Brain tumor segmentation, Tumor grading, Tumor classification.

1. INTRODUCTION

Brain tumor classification and segmentation are essential tasks in medical imaging, crucial for identifying and understanding brain tumours. These irregular cell masses, if left unchecked, can lead to serious consequences such as increased intracranial pressure, brain damage, or even death. To improve the accuracy and efficiency of tumor detection and segmentation, researchers have turned to deep learning techniques. Deep Learning models extract features from medical images and fuse them to enhance the classification accuracy of brain tumors. Additionally, researchers have explored

advanced architectures and methods, such as graph algorithms and generative adversarial networks, to improve accuracy. Despite these advancements, challenges remain in achieving accurate and reliable brain tumor classification, particularly in multi-class classification scenarios. By continuing to innovate and refine deep learning approaches, researchers hope to improve the outcomes for patients with brain tumors, providing more tailored and effective medical interventions. In summary, the classification and segmentation of brain tumors are vital for healthcare providers to diagnose patients accurately and plan their treatment

effectively. Existing works prone to sensitive to noise especially with a non-linear kernel, face scalability issues, computationally expensive and memory intensive. The proposed work overcome resource constraints using EfficientNetB3 and also unique mechanism enables ViT to comprehend the context of various parts of an image, ultimately resulting in more precise classification.

2. LITERATURE REVIEW

Atika Akter, Nazeela Nosheen, Sabbir Ahmed, Mariom Hossain, Mohammad Abu Yousuf, Mohammad Ali Abdullah Almoyad, Khondokar Fida Hasan, Mohammad Ali Moni[1] have proposed a Convolutional Neural Network (CNN) and U-Net based algorithms for classifying and segmenting brain MRI images to detect tumors. Four individual MRI datasets and two merged datasets are used to train and evaluate classification models, including a proposed CNN model and five pre-trained models. A U-Net model is also proposed and trained for MRI segmentation on a manually segmented dataset.

Guying Zhang, Jia Zhou, Guanghua He, Hancan Zhu[2] have proposed a new deep learning method for brain tumor segmentation from multi-modal MRI images using deep residual learning and multi-modality image feature fusion. It analyzes the BraTS2021 dataset and finds that different MRI modalities (T1, T1ce, T2, FLAIR) provide different and complimentary information about brain tumors. Directly concatenating the modalities ignores their unique characteristics. The proposed network adopts a U-Net architecture with an encoder-decoder structure. The proposed

method achieves state-of-the-art performance on the BraTS2021 dataset for brain tumor segmentation, demonstrating its effectiveness in leveraging multi-modal MRI information for accurate predictions.

Halit Çetiner, Sedat Metlek[3] have proposed a new deep learning model called DenseUNet+ for automated segmentation of brain tumors from multimodal MRI images. It aims to improve over existing UNet models by addressing issues like semantic gaps between the encoder-decoder sections. The commonly used MRI modalities as input for brain tumor segmentation are T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and fluid-attenuated inversion recovery (FLAIR). Each modality provides different pathological information: T1 distinguishes healthy tissues, T2 identifies edematous tumor regions, T1ce defines tumor borders, and FLAIR distinguishes between cerebrospinal fluid and edema.

3. METHODOLOGY

3.1 DATASET

Two datasets were used for both tumor classification and segmentation tasks. The classification dataset utilized consists of 7023 brain MRI images. These images are labelled with multiclass labels, and tumor type with annotated label. The different classes of tumor in the dataset are Meningitis, Pituitary, and Glioma tumors. The segmentation dataset contains 3929 brain MRI images where each pixel is annotated as tumor or non-tumor (background). Each image has a corresponding pixel-wise annotation mask indicating the tumor regions. The dataset covers various tumor types, sizes, and

locations to train a robust segmentation model.

3.2 DATA PRE-PROCESSING

Image pre-processing techniques such as normalization, resizing, and cropping are applied to ensure consistency and reduce computational complexity. For segmentation, the input images and masks are pre-processed together to ensure alignment between the image and its annotations.

3.3 CLASSIFICATION MODELS

Deep Learning models like EfficientNetB3, Vision Transformer and EfficientNetV2 are used for Classification of Brain MRI Images.

3.3.1 EFFICIENTNETB3

EfficientNetB3, as part of the EfficientNet family, is particularly notable for its approach to scaling. Rather than uniformly increasing the model's size, it employs a method of scaling that adjusts the depth, width, and resolution of the network in a balanced manner. This approach ensures that each dimension is scaled differently, optimizing the network architecture for improved performance without significantly increasing computational cost. EfficientNetB3's success lies in its ability to strike a balance between model size and accuracy, offering a practical solution for various computer vision tasks.

3.3.2 EFFICIENTNETV2

EfficientNetV2 represents a significant advancement in convolutional neural network architecture, offering faster training speeds and improved parameter efficiency compared to its predecessors. Developed through a combination of

training-aware neural architecture search and scaling techniques, EfficientNetV2 optimizes both training speed and parameter efficiency. Models within the EfficientNetV2 family are designed from a search space enriched with new operations, including Fused-MBConv, to enhance their effectiveness. Following a principle of systematic scaling in network depth and width, EfficientNetV2 prioritizes efficiency while maintaining high performance.

3.3.3 VISION TRANSFORMER

The Vision Transformer, or ViT, is a model for image classification that employs a Transformer-like architecture over patches of the image. In ViTs, images are transformed into sequences of tokens, breaking down the spatial relationships between pixels and treating each as a distinct entity. The transition from grid-based processing in Vision Transformers (ViTs) enables them to leverage self-attention mechanisms for understanding holistic context and extensive relationships. This capability empowers ViTs to discern intricate patterns and complex relationships within visual data.

3.4 SEGMENTATION MODELS

3.4.1 RESUNET

RESUNET, short for Deep Residual UNET, is a specialized architecture for semantic segmentation, aiming to achieve high performance with fewer parameters compared to UNET. It combines the strengths of both UNET and Deep Residual Learning. The architecture comprises an encoder, a bridging module, and a decoder. Encoder uses pre-activated residual blocks to process input images through three encoder blocks, extracting abstract features. The bridge connects the encoder and

decoder, facilitating feature sharing. The decoder receives feature maps from the bridge and skip connections from the encoder, gradually upsampling the features and reducing the number of channels.

3.4.2 ATTENTION UNET

Attention U-Net is an enhanced iteration of the U-Net framework, customized for tasks involving image segmentation. U-Net has gained prominence in medical imaging for its distinctive structure, featuring a contracting path for feature extraction and an expansive path for spatial information recovery. By integrating attention mechanisms, Attention U-Net enhances feature relevance while suppressing irrelevant ones, achieved through assigning attention weights to different image locations. This selective focus improves object localization accuracy, especially useful for small objects or those in complex backgrounds.

3.6 EVALUATION METRICS

3.6.1 LOSS FUNCTIONS

3.6.1.1 CROSS ENTROPY LOSS

Cross-entropy loss evaluates the performance of a classification model that outputs probabilities between 0 and 1. This loss increases as the predicted probability deviates from the actual label. This loss function is employed to adjust the weights of a machine learning model during training, aiming to minimize the error between the predicted and actual outcomes.

$$\text{logloss} = -\frac{1}{N} \sum_i^N \sum_j^M y_{ij} \log(p_{ij})$$

3.6.1.2 TVERSKY SCORE

The Tversky score serves as an asymmetric similarity measure between

two sets, often comparing a variant to a prototype. Specifically, it quantifies the dissimilarity between sets like predicted and ground truth binary masks.

$$TI = \frac{TP}{TP + \alpha FN + \beta FP}$$

3.6.1.3 FOCAL TVERSKY LOSS

The Focal Tversky Loss (FTL) is an extension of the Tversky loss that incorporates the gamma modifier from Focal Loss. This modification introduces a non-linear behavior to the loss function, allowing control over its behavior at different Tversky index values. The gamma parameter, denoted by γ , determines the degree of non-linearity in the loss function. When γ approaches positive infinity, the loss function's gradient also tends towards infinity as the Tversky Index (TI) approaches 1.

$$FTL = (1 - TI)^\gamma$$

3.6.2 DICE COEFFICIENT

The Dice coefficient, also referred to as the Sørensen-Dice coefficient, is a statistical measure employed to assess the similarity between two sets. It is used to evaluate algorithm performance by quantifying the overlap between the predicted segmentation mask and the ground truth mask. The coefficient's value ranges from 0 to 1.

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|}$$

3.6.3 IOU COEFFICIENT

The Intersection over Union (IoU) coefficient is a metric used to evaluate the

performance of image segmentation algorithms. It measures the similarity between the predicted segmentation mask and the ground truth mask by calculating the ratio of the intersection of the two masks to their union. The IoU coefficient ranges from 0 to 1.

$$\text{Jaccard} = \frac{|A \cap B|}{|A \cup B|}$$

5. RESULTS

5.1 CLASSIFICATION RESULTS

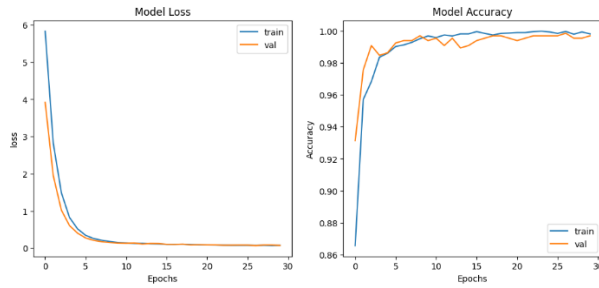


Fig 1: EfficientNetB3 Results

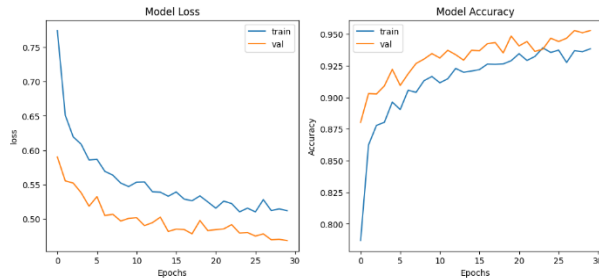


Fig 2: EfficientNetV2 Results

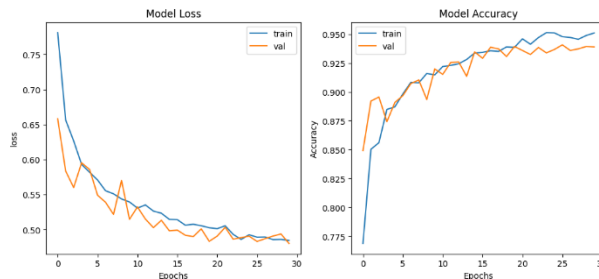


Fig 3: Vision Transformer Results

Figures 1, 2, 3 shows the cross-entropy loss and accuracy values of classification models used.

6. PERFORMANCE ANALYSIS

MODEL	FOCAL TVERSKY LOSS	TVERSKY SCORE	IOU COEFFICIENT	DICE COEFFICIENT
RESUNET	0.2007	0.8819	0.8243	0.8589
ATTENTION UNET	0.2879	0.8056	0.6577	0.7901

Table 1: Segmentation Models Metrics

Table 1 shows various evaluation metrics used for evaluating the Segmentation models used.

7. CONCLUSION

The utilization of Vision Transformer allows us to efficiently capture spatial relationships in the MRI scans, while EfficientNetB3 and EfficientNetV2 models help in extracting high-level features, both crucial for accurate classification. Classification models like EfficientNetB3, achieved an accuracy of nearly 99%, EfficientNetV2 achieved an accuracy of 95.2% and Vision Transformer achieved an accuracy of 95%. Segmentation models like ResUNET achieved a loss of nearly 0.2007. We plan to explore advanced attention mechanisms, such as self-attention and non-local attention. These mechanisms aid the model in emphasizing relevant features and enhancing its capacity to capture long-range dependencies in MRI scans

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